



Namal University, Mianwali

Software Requirements Specification (SRS)

Project: Point-of-Sale System

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1 introduction

1.1 Purpose

The purpose of this SRS is to formally define the requirements of the Point-of-Sale (POS) system for a pizza shop.

This document serves as a reference for developers, administrators, and end users to understand the system's intended functionality and constraints.

The SRS follows IEEE 830-1984 to ensure clarity, completeness, and consistency.

1.2 Scope

The POS system is the software product to be developed.

It shall support restaurant operations including order placement, deal recommendations, billing, kitchen coordination, and administrative management.

The system shall support multiple user roles: Waiter, Cashier, Kitchen Staff, and Admin. The goals of the system include:

- Improving order accuracy
- Reducing service time
- Providing real-time visibility of order status across all operational roles

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
POS	Point-of-Sale system used for sales and order processing
SRS	Software Requirements Specification
RP	Requirement Provider
Scrum	Agile development framework used for iterative development and team collaboration
Sprint	Time-boxed iteration in Scrum methodology

Table 1: Definitions, Acronyms, and Abbreviations

1.4 References

1. IEEE Std 830-1984: Software Requirements Specification, IEEE, 1984.
2. Team meeting with Requirement Provider (RP) regarding POS system requirements, December 2025.

1.5 Overview

This document provides a high-level description of the POS system.

General system characteristics, user roles, constraints, and assumptions are described in the following sections.

Specific functional and non-functional requirements, interface requirements, and system attributes are detailed in subsequent sections.

2 General Description

2.1 Product Perspective

The POS system is a centralized software application deployed on devices within the restaurant.

It replaces manual order handling and disconnected billing systems.

The system integrates order processing, kitchen notifications, and billing into a single platform with role-based user interfaces.

Supported hardware includes desktops, and tablets.

2.2 Product Functions

The system provides the following functions at a high level:

- Role-based login for Waiter, Cashier, Kitchen Staff, and Admin
- Menu display and order placement
- Automatic deal recommendation for cost optimization
- Real-time order status tracking
- Bill generation and payment handling
- Administrative reporting and menu management

2.3 User Characteristics

- **Waiter:** Places orders, selects tables, views order status, serves customers.
- **Cashier:** Generates bills, handles payments, views order summaries.
- **Kitchen Staff:** Receives and updates order preparation status.
- **Admin:** Manages menu items, views sales reports, and controls system data.

Users are assumed to have basic knowledge of POS systems.

2.4 General Constraints

- The system shall follow Scrum methodology for development.
- The SRS must comply with IEEE 830-1984 standards.
- Hardware and network limitations, interface compatibility, and security considerations shall be addressed in the Specific Requirements section.

2.5 Assumptions and Dependencies

- Users have basic POS knowledge.
- Required hardware devices (Desktops, tablets, printers, cash drawers) are available and functional.
- Payment gateways are provided by third-party services.
- Kitchen and cashier terminals are connected to the same network.

3 Specific Requirements

3.1 Functional Requirements

3.1.1 User Authentication

Functional Requirements:

- **FR-01.1** The system shall allow authorized users to log in using a unique username and password.
- **FR-01.2** The system shall validate the entered credentials against stored user records.
- **FR-01.3** The system shall grant access based on the authenticated user's role.
- **FR-01.4** The system shall deny access and display an error message if authentication fails.
- **FR-01.5** The system shall provide a "Forgot Password" feature to all users for password recovery.

Introduction: The system ensures that only authorized users can access the system and that users interact only with features relevant to their role.

Inputs:

- **Data:** Username and password.
- **Sources:** Login interface on devices (tablet, desktop).
- **Quantities:** One username and password pair per login.

- **Units of Measure:** Strings (username and password).
- **Timing:** On login attempt.
- **Valid Range:** Username and password strings of 8–16 alphabets, numeric and special characters.
- **Operator Controls:** User enters username and password and clicks “Login”.
- **Interface References:** GUI login screen design with forgot password option.

Processing:

- Verify input details against stored data.
- Check role and assign permissions after successful authentication.
- Deny access and display error message if invalid credentials.
- Process password recovery requests through forgot password feature.
- **Abnormal Situations:** Invalid login, communication failure with database, failed password recovery.
- **Parameters Affected:** User login status, role permissions.
- **Methods:** Check password securely using encryption.

Outputs:

- **Data:** Access granted or error message displayed.
- **Timing:** Immediate after login attempt.
- **Valid Range:** Success or failure states.
- **Illegal Values:** Display error message for invalid login details.
- **Interface References:** GUI for successful and unsuccessful login, password recovery screen.

3.1.2 Order Placement

Functional Requirements:

- **FR-02.1** The system shall allow the waiter to select a table number for a new order.
- **FR-02.2** The system shall display the available menu items to the waiter after table selection.
- **FR-02.3** The system shall allow the waiter to add one or more menu items to an order.

- **FR-02.4** The system shall allow the waiter to update or remove selected menu items before order submission.
- **FR-02.5** The system shall allow customers to modify, update, or cancel orders through the waiter screen before the waiter submits the order.
- **FR-02.6** The system shall automatically calculate a discount of 0.5% for each 1,000 PKR when the order total is greater than or equal to 10,000 PKR.
- **FR-02.7** The system shall automatically add complementary items when the order total is greater than or equal to 10,000 PKR, with total value up to 2% of the order amount.
- **FR-02.8** The system shall automatically select complementary items from a predefined list of small, low-cost menu items such as drinks, desserts, and appetizers. The total value of complementary items shall not exceed 2
- **FR-02.9** The system shall mark complementary items as free of cost.
- **FR-02.10** The system shall generate and display an order summary including ordered items, complementary items, subtotal, discount, and total amount upon confirmation.
- **FR-02.11** The system shall forward the confirmed order to the cashier and kitchen staff.

Introduction: The software allows the waiter to take customer orders accurately, select table numbers, add menu items, apply automatic discounts and complementary items based on order value.

Inputs:

- **Data:** Table number, menu items, quantities.
- **Sources:** Waiter device interface.
- **Quantities:** One table number, multiple items per order.
- **Units of Measure:** Integer (table number, quantity), currency (item price, PKR).
- **Timing:** During customer ordering.
- **Valid Range:** Table number 1–50; quantity > 0 ; item prices > 0 ; order total ≥ 0 .
- **Operator Controls:** Touchscreen to select table, items, quantities, and confirm order.
- **Interface References:** GUI menu screen, order summary screen.

Processing:

- Display menu after table selection.
- Add, update, or remove items in order before submission.

- Calculate order total continuously.
- Apply 0.5% discount per 1,000 PKR when total $\geq 10,000$ PKR.
- Automatically select and add complementary items (up to 2% of order value) when total $\geq 10,000$ PKR.
- Validate item selection and quantity.
- Confirm and submit order.
- **Abnormal Situations:** Item unavailable, quantity exceeds stock, communication failure, no complementary items available.
- **Parameters Affected:** Order contents, subtotal, discount amount, complementary items, total price, status.
- **Methods:** Arithmetic calculations for totals, discounts, and complementary item values.

Outputs:

- **Data:** Order summary with table number, items, complementary items, quantities, subtotal, discount, total amount.
- **Destinations:** Waiter, cashier and kitchen staff devices.
- **Timing:** Immediately after confirmation.
- **Valid Range:** Order totals ≥ 0 ; discount ≥ 0 ; complementary item value $\leq 2\%$ of order amount.
- **Illegal Values:** Display error if item or quantity invalid.
- **Interface References:** GUI order summary screen showing all items, complementary items, discount, and totals.

3.1.3 Order Processing

Functional Requirements:

- **FR-03.1** The system shall assign a unique order ID to each confirmed order.
- **FR-03.2** The system shall transmit the confirmed order details including complementary items and discount to the cashier and kitchen staff.
- **FR-03.3** The system shall set the initial status of a new order to *Order Pending*.
- **FR-03.4** The system shall allow customers to modify or cancel orders before kitchen staff accepts the order.

- **FR-03.5** The system shall prevent modification or cancellation of existing items after kitchen staff accepts the order.
- **FR-03.6** The system shall allow additional items to be added to an order after kitchen acceptance.
- **FR-03.7** The system shall track and update the order status throughout its lifecycle: Order Pending → In Preparation → Prepared → Served → Completed.
- **FR-03.8** The system shall display updated order status information to the waiter, cashier, and kitchen staff.

Introduction: The software manages the complete order lifecycle from creation through completion, sends orders to the cashier and kitchen, and tracks their status while enforcing modification and cancellation rules.

Inputs:

- **Data:** Confirmed order (items, complementary items, quantities, discount, table number), modification requests, cancellation requests.
- **Sources:** Waiter device.
- **Quantities:** One order per table.
- **Units of Measure:** String (items), integer (quantity), currency (price, discount, PKR).
- **Timing:** Immediately after order submission and throughout order lifecycle: Order Pending → In Preparation → Prepared → Served → Completed.
- **Valid Range:** Order list cannot be empty; valid status values.
- **Operator Controls:** Waiter controls for modifications and additions through touchscreen.
- **Interface References:** Order modification interface, status display screens, popups for kitchen staff and order display for cashier.

Processing:

- Assign unique order ID.
- Send order summary to cashier and kitchen staff.
- Set initial order status to Order Pending.
- Allow modifications and cancellations before kitchen acceptance.
- Lock existing items from modification/cancellation after kitchen acceptance.
- Allow additional items to be added after kitchen acceptance.

- Update status in real time through lifecycle.
- **Abnormal Situations:** Network failure, duplicate order ID, invalid modification request.
- **Parameters Affected:** Order status, order ID, order contents, modification timestamp.
- **Methods:** Update order status automatically, assign unique IDs, validate modification rules.

Outputs:

- **Data:** Order summary, status updates, modification confirmations.
- **Destinations:** Cashier and kitchen staff devices, waiter device.
- **Timing:** Real time.
- **Valid Range:** Status: Order Pending, In Preparation, Prepared, Served, Completed.
- **Illegal Values:** Error notifications if order has an invalid status or invalid modification attempted.
- **Interface References:** Dashboard and notification GUI for cashier, kitchen staff, and waiter.

3.1.4 Kitchen Operations

Functional Requirements:

- **FR-04.1** The system shall display all pending orders to the kitchen staff.
- **FR-04.2** The system shall allow kitchen staff to accept an assigned order for preparation.
- **FR-04.3** The system shall update the order status to In Preparation when kitchen staff accepts the order.
- **FR-04.4** The system shall prevent order modifications and cancellations of existing items once kitchen staff accepts the order.
- **FR-04.5** The system shall allow kitchen staff to mark an order as Prepared upon completion.
- **FR-04.6** The system shall notify the corresponding waiter when an order is marked as Prepared.

Introduction: The software supports kitchen staff in managing order workflow, preparing orders including complementary items, and notifying the waiter upon completion.

Inputs:

- **Data:** Pending orders, acceptance confirmations, completion updates.
- **Sources:** Kitchen staff device.
- **Quantities:** One or multiple orders displayed to kitchen staff.
- **Units of Measure:** Strings (status), integers (order ID).
- **Timing:** During order preparation.
- **Valid Range:** Status values: Order Pending → In Preparation → Prepared.
- **Operator Controls:** Touchscreen or desktop to accept or complete orders.
- **Interface References:** Kitchen dashboard GUI.

Processing:

- Display pending orders with all items including complementary items.
- Update order status on acceptance to In Preparation.
- Lock order from customer modifications and cancellations upon acceptance.
- Update order status to Prepared upon completion.
- Notify corresponding waiter when order is completed.
- **Abnormal Situations:** System crash, missing order ID, notification failure.
- **Parameters Affected:** Order status, notifications, modification locks.
- **Methods:** Update order status automatically, send notifications.

Outputs:

- **Data:** Updated order status, waiter notifications.
- **Destinations:** Kitchen dashboard, waiter interface.
- **Timing:** Real-time.
- **Valid Range:** Status: Order Pending → In Preparation → Prepared.
- **Illegal Values:** Error displayed on dashboard if status is invalid.
- **Interface References:** Dashboard and notification popup GUI.

3.1.5 Order Serving

Functional Requirements:

- **FR-05.1** The system shall notify the waiter when an order is marked as Prepared by the kitchen staff.
- **FR-05.2** The system shall allow the waiter to mark an order as Served to the customer.
- **FR-05.3** The system shall update the order status to Served when the waiter confirms serving.
- **FR-05.4** The system shall allow the waiter to mark an order as Completed after the customer finishes the meal.
- **FR-05.5** The system shall update the order status to Completed and make it visible to cashier and admin dashboards.

Introduction: The software tracks the serving of orders and updates the status after the order is served to the customer and consumed.

Inputs:

- **Data:** Order prepared notification from kitchen.
- **Sources:** Kitchen staff device (Prepared) and waiter device.
- **Quantities:** One order per table.
- **Units of Measure:** Strings (status).
- **Timing:** After preparation and during serving.
- **Valid Range:** Status values: Prepared → Served → Completed.
- **Operator Controls:** Touchscreen or keyboard to mark order as Served or Completed.
- **Interface References:** Waiter order screen GUI.

Processing:

- Notify waiter when order is prepared and ready.
- Waiter serves the order and updates status to Served.
- Update status to Completed after customer finishes.
- **Abnormal Situations:** Missed notification, incorrect table selection.
- **Parameters Affected:** Order status, completion timestamp.
- **Methods:** Real-time status updates and notifications.

Outputs:

- **Data:** Updated order status.
- **Destinations:** Waiter, cashier, and admin dashboards.
- **Timing:** Real-time.
- **Valid Range:** Status: Prepared → Served → Completed.
- **Illegal Values:** Display error to waiter for invalid status transition.
- **Interface References:** Status update GUI.

3.1.6 Billing and Payment

Functional Requirements:

- **FR-06.1** The system shall generate a bill for each completed order including item prices, complementary items (automatically added when order total $\geq 10,000$ PKR, total value up to 2% of the order amount), subtotal, discount (0.5% per 1,000 PKR of order total if $\geq 10,000$ PKR), taxes, service charges, and final payable amount.
- **FR-06.2** The system shall allow the cashier to select a payment method: cash, card swipe, or online payment via bank or digital wallet (e.g., JazzCash).
- **FR-06.3** The system shall validate payments based on the selected method: for cash, the entered amount must be $\geq$ bill total; for card swipe, the system shall send a request to the bank/payment processor for authorization; for online payment, the system shall send a request to the payment gateway or bank API and confirm successful payment before updating order status.
- **FR-06.4** The system shall process payments securely, update the payment status in real-time, and log transaction details for accounting and auditing.
- **FR-06.5** The system shall generate a payment receipt containing receipt number, order ID, date and time, table number, ordered items, complementary items, subtotal, discount, taxes, final payable amount, payment method, and payment status, and display it to the customer and cashier.
- **FR-06.6** The system shall store all payment receipts and transaction details securely for accounting, auditing, and reporting purposes.
- **FR-06.7** After successful payment, the system shall allow the customer to submit feedback with a rating from 1 to 5.

Introduction: The software handles bill generation and payment processing, supporting cash, card swipe, and online payments, ensuring accurate calculations including discounts and complementary items, and secure transactions.

Inputs:

- Data: Order summary (items, complementary items, quantities, prices, discount), selected payment method, payment amount provided by customer
- Sources: Cashier device or online payment interface
- Quantities: One payment per completed order
- Units of Measure: Currency (PKR)
- Timing: After order completion
- Valid Range: Payment amount $\geq$ final payable amount including taxes and discounts
- Operator Controls: Cash entry, card swipe, or online payment confirmation via GUI
- Interface References: Billing and payment screen GUI

Processing:

- Calculate subtotal from ordered items
- Apply discount (0.5% per 1,000 PKR if total $\geq$ 10,000 PKR)
- Add complementary items at zero cost
- Calculate taxes and service charges
- Calculate final payable amount
- Validate the payment amount against the final total
- Process payment through the selected method and update payment status
- Generate receipt with all required details
- Store transaction records securely for accounting and auditing
- Abnormal Situations: Insufficient payment, network failure during online payment, transaction declined, card swipe failure
- Parameters Affected: Payment status, order records, transaction logs
- Methods: Arithmetic calculations for totals, payment validation, secure transaction processing

Outputs:

- Data: Bill, payment confirmation receipt with receipt number, order ID, date and time, table number, items, complementary items, subtotal, discount, taxes, final amount, payment method, and payment status
- Destinations: Cashier interface, customer (printed receipt), admin dashboard for records

- **Timing:** Immediately after successful payment
- **Valid Range:** Final payable amount > 0 ; discount ≥ 0
- **Illegal Values:** Error messages for invalid payment or system failures
- **Interface References:** Billing, payment, and receipt GUI screens

3.1.7 Customer Feedback

Functional Requirements:

- **FR-07.1** The system shall allow customers to submit feedback after successful payment.
- **FR-07.2** The system shall provide a rating scale from 1 to 5 for customer feedback, where 1 = Very Poor, 2 = Poor, 3 = Average, 4 = Good, and 5 = Excellent.
- **FR-07.3** The system shall store customer feedback with associated order information for analysis.

Introduction: The software collects customer satisfaction feedback to help improve service quality.

Inputs:

- **Data:** Customer rating (1–5), optional comments.
- **Sources:** Cashier device or customer-facing interface.
- **Quantities:** One feedback submission per completed payment.
- **Units of Measure:** Integer (rating 1–5), string (comments).
- **Timing:** After successful payment completion.
- **Valid Range:** Rating: 1, 2, 3, 4, or 5.
- **Operator Controls:** Touchscreen or button interface for rating selection.
- **Interface References:** Feedback submission GUI.

Processing:

- Display feedback interface after payment confirmation.
- Validate rating input (1–5).
- Associate feedback with order ID and timestamp.
- Store feedback in database.
- **Abnormal Situations:** Database connection failure, invalid rating value.

- **Parameters Affected:** Feedback records, customer satisfaction data.
- **Methods:** Data validation and secure storage.

Outputs:

- **Data:** Stored feedback with order ID, rating, timestamp, and optional comments.
- **Destinations:** Admin dashboard for analysis.
- **Timing:** Immediately after submission.
- **Valid Range:** Rating: 1–5.
- **Illegal Values:** Error message for invalid rating.
- **Interface References:** Feedback confirmation screen.

3.1.8 Administration

Functional Requirements:

- **FR-08.1** The system shall generate detailed sales reports (daily, weekly, monthly) upon admin request or schedule.
- **FR-08.2** The system shall include in daily reports: number of orders prepared, total sales, total discounts given, complementary item value, and payment summaries (cash, card, online).
- **FR-08.3** The system shall identify and display most-selling and low-performing menu items for analysis.
- **FR-08.4** The system shall allow the admin to add, update, or remove menu items.
- **FR-08.5** The system shall allow the admin to manage the predefined complementary items list.
- **FR-08.6** The system shall allow the admin to manage user accounts, including adding, updating, deactivating, or removing users.
- **FR-08.7** The system shall display error messages for invalid input, duplicate entries, or failed operations.

Introduction: The software allows admin to monitor restaurant performance through comprehensive reports, manage menu items including complementary items, and control user accounts.

Inputs:

- **Data:** Requests for sales reports, menu updates, complementary items management, and user management.

- **Sources:** Admin interface.
- **Quantities:** One action per request (report generation, menu modification, or user account management).
- **Units of Measure:** String (menu items, reports), currency (sales, discounts, PKR), integer (order count).
- **Timing:** On demand or scheduled intervals.
- **Valid Range:** Sales ≥ 0 ; discount ≥ 0 ; order count ≥ 0 .
- **Operator Controls:** Admin clicks and form inputs.
- **Interface References:** Admin dashboard GUI.

Processing:

- Generate detailed sales reports (daily, weekly, monthly).
- Calculate and display: total orders, total sales, total discounts, complementary item values, payment method breakdowns.
- Identify most-selling items and low-performing items for analysis.
- Add, update, or remove menu items.
- Manage complementary items list.
- Manage user accounts: add, update, deactivate, or remove users.
- **Abnormal Situations:** Invalid input data, system crash, duplicate menu entries, report generation failure.
- **Parameters Affected:** Menu database, complementary items database, sales report data, user account records.
- **Methods:** Data aggregation, statistical analysis, database operations.

Outputs:

- **Data:** Generated reports with order count, sales totals, discount totals, complementary item values, payment summaries; updated menu items; updated complementary items; user account changes.
- **Destinations:** Admin dashboard, waiter menu interface.
- **Timing:** Real time updates for changes, scheduled outputs for reports.
- **Valid Range:** Accurate sales details, valid menu item details, correct user account details.
- **Illegal Values:** Error messages displayed for invalid input, failed operations, or duplicates.
- **Interface References:** Admin dashboard GUI, report generation screens.

3.2 Performance Requirements

3.2.1 Static Numerical Requirements

- The system shall support at least 50 concurrent users (waiters, cashiers, kitchen staff, admins).
- The system shall handle 50 simultaneous users without performance degradation.
- The system shall support processing up to 500 orders per day.
- The system shall maintain 99.99% consistency of all order, payment, discount, complementary item, and report records for up to 500 daily transactions, verified by automated integrity checks at least once per hour.

3.2.2 Dynamic Numerical Requirements

- The system shall load menu within 3 seconds.
- The system shall calculate discounts and select complementary items within 1 second of order total calculation.
- The system shall process orders entered by the waiter and update the status to the cashier and kitchen staff within 2 seconds per order.
- The system shall reflect order status updates in all relevant user interfaces in real time with a maximum latency of 1 second.
- The system shall generate reports for daily sales within 5 seconds and for up to one year of data within 10 seconds.
- The system shall complete and log payment transactions successfully at least 99.99% of the time.
- The system shall allow new users to learn and perform primary tasks (login, place/update orders, generate reports) within 10 minutes of training, for at least 90% of users tested.
- The system shall provide visual notifications (pop-ups) for order updates within 1 second of the triggering event.

3.3 Design Constraints

3.3.1 Standards Compliance

- The system shall follow standard rules for report formats, data naming, and review records.
- The system shall record all changes in orders, payments, or reports with before and after values, and these records shall be kept for at least 12 months.

- The system shall produce reports within 5 seconds for daily reports and within 10 seconds for up to 1 year when requested.
- Payment receipts shall include all required fields: receipt number, order ID, date and time, table number, ordered items, complementary items, subtotal, discount, taxes, final payable amount, payment method, and payment status.

3.3.2 Hardware Limitations

- The system shall run on Windows 10 or newer and Android 10 or newer.
- The system shall work on PCs with at least 4 GB RAM, 2-core CPU, and 256 GB storage, and on Android devices with at least 2 GB RAM and 16 GB storage.
- The system shall process all orders, payments, discount calculations, and reports without using more than 80% of CPU or memory at peak load.
- The system shall work properly with network speeds of at least 10 Mbps for proper operation across multiple devices.

3.4 Attributes

3.4.1 Availability

- The system shall be 99.99% available during restaurant hours (for example 10:00 AM–11:00 PM).
- The system shall have a maximum downtime of 2 hours per month for maintenance.
- The system shall maintain 99.5% uptime for critical services (login, order processing, payment) during operating hours.

3.4.2 Security

- Each user shall access the system only after successful authentication.
- Admin operations shall require multi-factor authentication.
- The system shall log and report unauthorized access attempts within 10 seconds.
- Payment processing shall use secure encryption protocols.

3.4.3 Maintainability

- The system shall reflect updates to menu items, complementary items, or discount policies within 5 seconds of admin confirmation.
- The system shall allow incremental updates for new features without downtime exceeding 10 minutes.

3.4.4 Portability / Transferability

- The system shall operate on Windows 10 or newer and Android 10 or newer.
- The system shall maintain consistent user interface and functionality across supported platforms.

3.4.5 Data Integrity

- The system shall maintain 100% consistency of order, payment, and report data for up to 500 daily transactions.
- In case of power failure or system crash, the system shall restore the last consistent state within 30 seconds of reboot.

3.5 External Interface Requirements

3.5.1 User Interfaces

- The system shall provide separate graphical user interfaces for Waiter, Cashier, Kitchen Staff, and Admin.
- Interfaces shall allow login, order placement, order tracking, bill generation, and reporting according to user roles.
- The GUI shall include touch friendly elements for android use.
- Interfaces shall provide visual notifications (pop-ups) for order updates, errors, and confirmations within 1 second of the event.
- Each interface shall use consistent screen formats and layouts.
- New users shall be able to perform primary tasks (login, place/update orders, generate reports) within 10 minutes of training, verified with at least 90% success rate in test users.

3.5.2 Hardware Interfaces

- The system shall run on desktop PCs, and Android.
- The system shall interface with receipt printers for printing bills and order receipts.
- Cash drawers shall open automatically after successful cash payment.
- The system shall support standard screen resolutions of at least 1280×720 for PCs and tablets.

3.5.3 Software Interfaces

- The system shall integrate with the restaurant database for storing user accounts, orders, payments, and menu items.
- The system shall integrate with payment gateways (online payment APIs) to process digital payments securely.

3.5.4 Communications Interfaces

- The system shall communicate over the local area network (LAN) to synchronize orders across devices in real time.
- Internet connectivity shall be used for online payment processing and admin access.

3.6 Other Requirements

3.6.1 Database Requirements

- The database shall store orders, payments, user accounts, and menu items.
- Data shall be accessed quickly so that order status and reports are available in real time.
- The system shall let at least 50 users access and update data at the same time without errors or delays.
- Data elements and records shall follow defined field types, lengths, and relationships.
- The database shall be organized to handle up to 500 transactions per day without errors or performance problems.
- Dynamic organization: system shall maintain real-time updates for orders, payments, and reports.
- Data shall be retained for at least 12 months until admin specified.
- Cash drawers shall open automatically after successful cash payment.

A Use Case Diagram

B Context Diagram



Figure 1: Use Case Diagram of the POS System



Figure 2: Context Diagram of the POS System